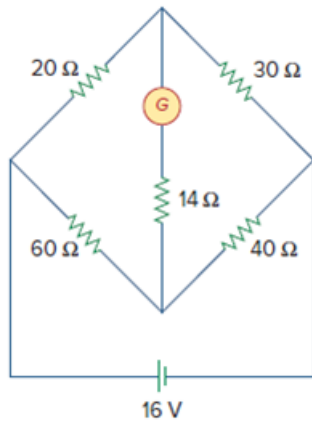


Problem

Obtain the current through the galvanometer, having a resistance of $14\ \Omega$, in the Wheatstone bridge shown in Fig. 4.66.

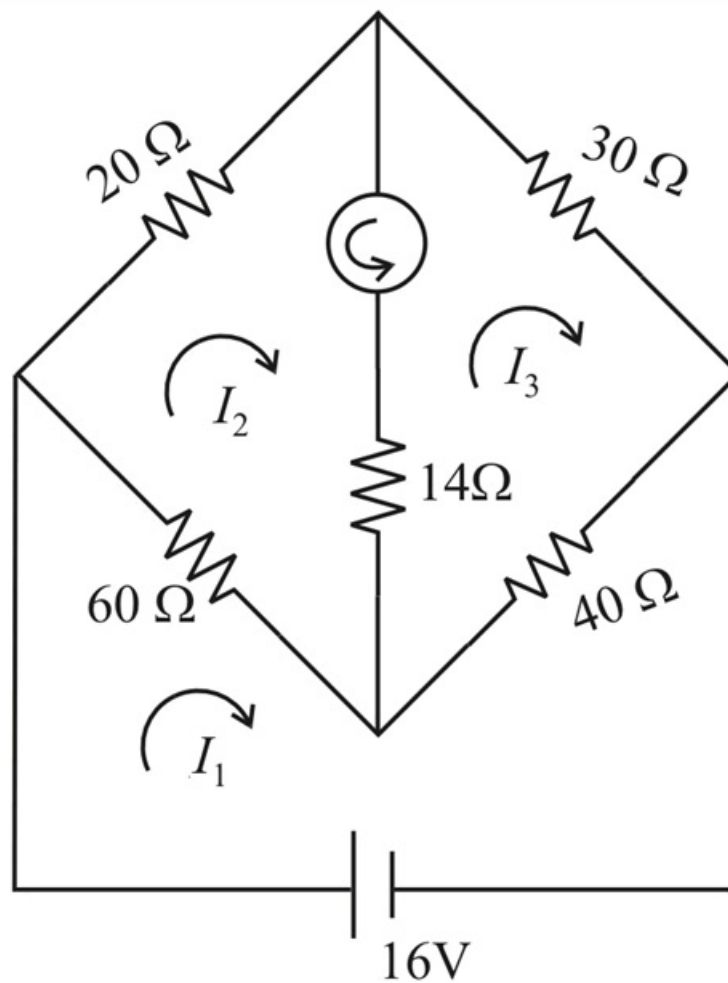
Figure. 4.66:



Step-by-step solution

Step 1 of 4

Considering the galvanometer



Step 2 of 4

By KVL we have the equation

$$-16 + 60(I_1 - I_2) + 40(I_1 - I_3) = 0$$

$$-16 + 60I_1 - 60I_2 + 40I_1 - 40I_3 = 0$$

$$100I_1 - 60I_2 - 40I_3 = 16 \quad \text{.....(1)}$$

$$60(I_2 - I_1) + 20I_2 + 14(I_2 - I_3) = 0$$

$$60I_2 - 60I_1 + 20I_2 + 14I_2 - 14I_3 = 0$$

$$-60I_1 + 94I_2 - 14I_3 = 0 \quad \text{.....(2)}$$

Step 3 of 4

$$14(I_3 - I_2) + 30I_3 + 40(I_3 - I_1) = 0$$

$$14I_3 - 14I_2 + 30I_3 + 40I_3 - 40I_1 = 0$$

$$-40I_1 - 14I_2 + 84I_3 = 0 \quad \text{.....(3)}$$

Step 4 of 4

Solving (1) and (2) and (3) equations

$$I_1 = \frac{308}{625}$$

$$= 0.4928$$

$$I_2 = \frac{224}{625}$$

$$= 0.3584$$

$$I_3 = \frac{184}{625}$$

$$= 0.2944$$

Current through galvanometer is

$$I_g = I_2 - I_3$$

$$I_g = 0.3584 - 0.2944$$

$$I_g = 0.064$$

$$I_g = 64 \times 10^{-3} \text{ A}$$

$$\boxed{I_g = 64 \text{ mA}}$$